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Electric Field Practice Questions

1. An electron is sitting in an electric field that points left and has a magnitude of 5.2 N/C . Find the magnitude and direction of the force acting on the electron.

2. What is the electric field 3 mm away from a charge of +5 mC?

Name: _____

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Period: _____

3. A charge of $q_1 = +40$ nC sits 5 mm away from a charge of $q_2 = -50$ nC. (*Hint:* 1 nC = 10^{-9} C)

- (a) Draw the electric field lines.
- (b) What is the magnitude and direction of the electric field halfway between the two charges?
- (c) What is the magnitude and direction of the electric field at $x = 8$ mm?

